

Classification of Self-Actions and the Unique Involutive Survivor

Abstract

We present two eliminative classifications over one kernel. The first classifies self-actions $a : \alpha \rightarrow \alpha$ by local orbit shape. Rest, asymmetric step, and erasure each meet a formal obstruction in the ambient of pointed symmetric steps; the surviving shape is the fixed-point-free two-cycle. That survivor is characterised by a universal property: $(\text{Bool}, \neg)$ is initial among pointed unoriented acts, and every fundamental act reads as negation. From fundamentality we obtain a Lawvere-style seal with quantitative grain (names lag predicates at every level, and the lag recedes under completion), an ω -ladder of namers whose colimit is duration, and, after transferring losslessness to forward microsteps by two-bounce factorisation, registration, capacity, and a single $\mathbb{Z}/2\mathbb{Z}$ -underdetermination with three faces. Canonicity of the two-bounce pair fails; channel *swap* nevertheless covers step reversal, and content gauges are separated from packaging gauges so that factorisation non-uniqueness does not inflate the dictionary $\mathbb{Z}/2\mathbb{Z}$. The second classification repeats the eliminative method on motion: among second-order nearest-neighbour updates over the same kernel, losslessness forces the xor form, and parity, pole invariance, and a drawing condition leave exactly one coupling up to pointwise flip, whose dynamics is the discrete wave step. A carrier-size dichotomy places the residual flip: on carriers divisible by four it is packaging, realised by an explicit conjugating mask, and on every other carrier no equivariant identification of the two dynamics exists, so there the flip carries content. The classified dynamics factors into two involutions definitionally, reverses by swapping them, carries a period spectrum governed by a gcd law, and returns within capacity. At the edge of the ladder the complete gauge histories form a boundary: every rung embeds there with effective separation, the seal persists at the boundary's own carrier, and the exclusion of erasure fails, so losslessness is a finite-stage law whose failure at the colimit is itself a theorem. On this base we assemble a conditional monist package, *self-articulation*: a registered denial of one drawn distinction draws the registering carrier (retorsion); a drawn distinction forces the live involutive reading; witnesses arise as ladder inhabitants; every presentation of the canon to a second term retracts the canon's own articulation; and completeness exists only as duration. The dictionary carries two worked exhibits: a certificate for the meta-problem of consciousness, deriving the report structure of the explanatory gap while refusing all qualia claims, and a certificate identifying the classified dynamics with reversible lattice computation. The development is machine-checked in Lean 4 (no Mathlib, no sorry).

Keywords. classification; involution; universal property; Lawvere diagonal; wave dynamics; reversible computation; self-articulation; retorsion; meta-problem; underdetermination; gauge; machine-checked mathematics.

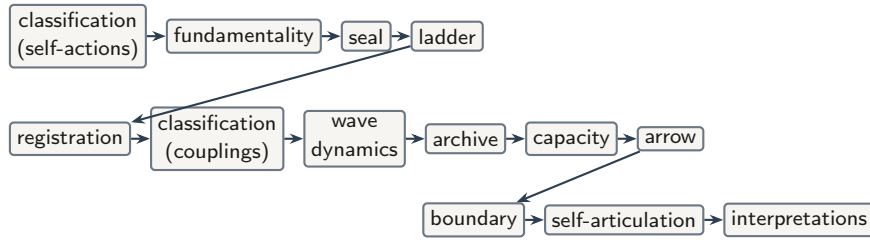
1 Introduction

The central contribution of this paper is a twofold eliminative classification: the kernel is forced, and its motion is forced. The static half classifies self-actions. Given $a : \alpha \rightarrow \alpha$, the local possibilities at a state are rest, two-cycle, asymmetric step, and—as a pairwise defect—erasure. In a precisely specified ambient, every shape except one encounters an obstruction. The survivor is the live two-cycle. Negation on Bool [5] is not assumed at the outset; it is the unique object (up to label swap) satisfying the universal property, in the standard categorical sense [21], of that survivor.

The dynamic half repeats the same eliminative method one level up, on the equation of motion. Among second-order nearest-neighbour updates over the kernel, the demand that no two histories merge forces the update into an xor form, and three ambient conditions that mirror the static ones (parity, pole invariance, and a drawing condition) leave exactly one coupling up to pointwise flip. The survivor’s dynamics is the discrete wave step: it factors into two involutions definitionally, it reverses by swapping them, its period spectrum obeys a gcd law and is reshaped by boundary conditions alone, and every state returns within the capacity of its carrier. Motion is thereby classified with the same discipline as the kernel, and no equation of motion is chosen by hand.

A further contribution assembles, on that base, a conditional monist package. Given one drawn distinction, articulation as negation is forced, witnesses arise only as inhabitants of the resulting ladder, no presentation of the canon to a second term carries content beyond the canon’s own articulation, and completeness exists only as the colimit of ascent. The package is a single machine-checked conjunction (Theorem 14.7); its identification with worldhood is confined to a marked interpretation, and the dictionary works two certificates, one for the meta-problem of consciousness and one identifying the classified dynamics with reversible lattice computation.

The formal development then follows a fixed dependency:



Each arrow is a theorem or a defined interface; the second classification node feeds the dynamics row, and philosophical commentary is postponed until the supporting machinery is in place.

Method and layers. The text maintains three layers.

- (1) **Formal mathematics** — definitions, theorems, proofs.
- (2) **Philosophical interpretation** — what the mathematics suggests, marked as such.
- (3) **Dictionary** — optional physical or contemplative identifications, admitted only by certificate.

Claims are tagged by status: proved, corollary, interpretation, open, refused, or philosophical residue. Residue is reduced by reclassification: items become theorems, audited residues at the base, or certificate data (Section 16). The companion Lean 4 corpus [10] (no Mathlib [23], no sorry) supplies declaration names; this paper is self-contained at the level of statements.

Roadmap. Section 2 gives the architecture. Sections 3–4 carry out the static classification and the universal characterisation of the survivor. Sections 5–6 obtain seal and ladder. Section 7 transfers losslessness by two-bounce factorisation. Section 8 carries out the dynamic classification, and Section 9 develops the classified wave dynamics. Sections 10–11 record registration consequences, capacity, and the arrow as address. Section 12 states the single $\mathbb{Z}/2\mathbb{Z}$, the gauge demarcation, and the placement of the coupling flip. Section 13 presents the boundary of complete gauge histories and the edge failure of losslessness. Section 14 assembles the self-articulation package and sorts its admissible readings. Section 15 confines the dictionary to two worked certificates. Section 16 collects open, refused, and philosophical items. Section 17 situates the development in prior work. Section 18 concludes.

2 Architecture

The companion corpus is stratified into three layers with downward-only imports (Figure 1).

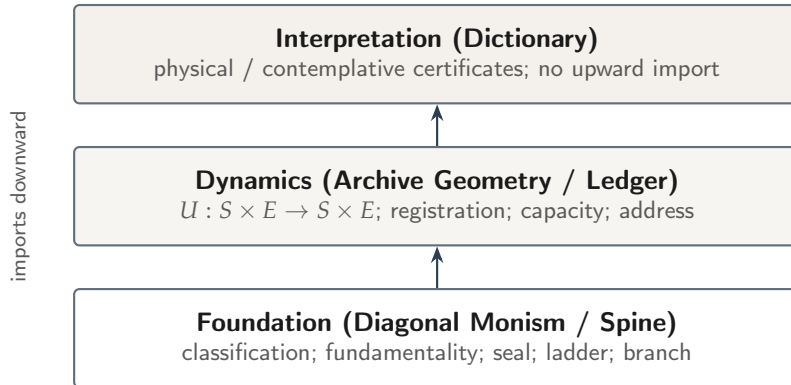


Figure 1: Three-layer architecture. Arrows indicate permitted dependence (foundation supports dynamics; dynamics support interpretation). The figure orients the reader; it proves nothing.

Within the foundation, the dependency is classification $\rightarrow$ fundamentality $\rightarrow$ seal $\rightarrow$ ladder. The dynamics layer imports losslessness along an interface (registration via two-bounce), classifies the couplings and develops the resulting wave dynamics, realises archive and capacity on the ladder carrier, and exports address as the arrow reading. The dictionary layer consumes those exports; it does not feed them.

The companion corpus realises the three layers as four build packages: a spine package (foundation); a ledger package and a bridge package jointly constituting the dynamics layer, the bridge holding the interface theorems that import both spine and ledger together with the dynamics arc of Sections 8–9 (modules `DynamicsAmbient`, `WaveDynamics`, `PeriodSpectrum`, `Recurrence`); and a dictionary package with exhibits (interpretation). Imports remain downward throughout. A fifth package, `formal/experimental`, is quaran-

ted by construction: no spine, ledger, bridge, or dictionary module imports it, and the annex at the end of the paper inventories what it holds. Citations at that package occur in exactly two places, the boundary results of Section 13 and the coupling-flip placement of Section 12, and they are flagged as such where they appear. A few dynamics-layer declarations bear physically suggestive names as mnemonics for combinatorial statements; the names import no identification, which remains certificate-only (Section 15).

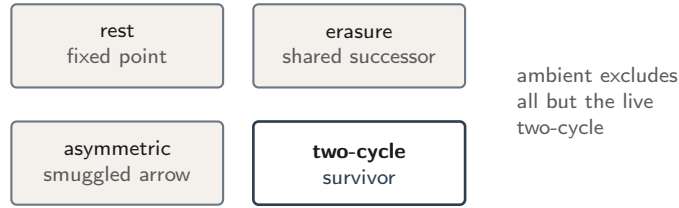
3 Classification of self-actions

Definition 3.1 (Arena data). Fix a type α and a map $a : \alpha \rightarrow \alpha$. Write

$$\text{Step}(a)(x, y) :\Leftrightarrow y = a(x).$$

Call a a *symmetric step* when $\text{Step}(a)$ is a symmetric relation; equivalently, $a \circ a = \text{id}$. Call a *live* when $\forall x. a(x) \neq x$. Call a *lossless* when a is injective. Call a *erasing* when distinct states share a successor.

Theorem 3.2 (Pointwise classification). *For decidable equality on α , at each z exactly one of the following holds: (i) rest ($a(z) = z$), with all iterates stagnant; (ii) two-cycle at z ($a(z) \neq z$ and $a(a(z)) = z$), with every iterate equal to z or $a(z)$; (iii) asymmetric step at z ($a(a(z)) \neq z$). The three cases are pairwise incompatible. Erasure is diagnosed separately as a pairwise defect of losslessness.*



Remark 3.3 (Status). Theorem 3.2 is proved. It is the eliminative engine of the paper: subsequent sections impose ambient constraints under which (i) and (iii), and erasure, are excluded as candidates for fundamentality.

Symmetric step is the ambient equation used below. It jointly excludes asymmetric arrow (case (iii) globally) and, by the next corollary, erasure.

Proposition 3.4 (In-arena losslessness). *If a is a symmetric step, then a is not erasing.*

Remark 3.5 (Status). Corollary of Definition 3.1 (involution $\Rightarrow$ injectivity). Not an independent creed.

Theorem 3.6 (Ambient uniqueness, one-variable equational fragment). *Relative to the arena signature of Definition 3.1 (one type, one endomap), every single equation in one free variable has the form $a^m(x) = a^n(x)$. Among those equations:*

- (i) $a^m = a^n$ with $n \geq 1$ need not force injectivity ($a^2 = a$ admits erasure);
- (ii) $a^m = \text{id}$ forces losslessness; $m = 1$ is rest; for every $m \geq 3$ an m -cycle is a countermodel with an asymmetric point;
- (iii) hence $a^2 = \text{id}$ is the unique nontrivial one-variable single equation that excludes both smuggled arrow and smuggled loss.

Remark 3.7 (Status). Proved by exhaustive classification of one-variable equations relative to the signature, with finite countermodels (including the m -cycle family for all $m \geq 3$). Multi-variable single equations over the same signature—for example $\forall x \forall y. a(x) = a(y)$ —force constancy on the image and are erasing on any carrier with two or more points, so they die immediately as ambients for fundamentality. Ambients beyond the one-variable equational fragment are refused by parsimony of the arena language (Section 16). Interpretation 3.8 records only the residual choice of signature itself.

Interpretation 3.8. Symmetric step is not an arbitrary modelling confession: by Theorem 3.6 it is the unique nontrivial one-variable single-equation ambient in the arena signature that bans smuggled time and smuggled loss. What remains is the signature declaration of Definition 3.1—one type, one endomap—stated in daylight, not residue.

4 Fundamentality as universal property

Definition 4.1 (Pointed unoriented acts). Objects are triples (α, a, z) with a a symmetric step and $z : \alpha$ a seed. Morphisms are pointed equivariant maps.

Theorem 4.2 (Orbit liveness as non-degeneracy). *Let a be a symmetric step and z a seed. The orbit of z is either the singleton $\{z\}$ or the pair $\{z, a(z)\}$. Liveness on that orbit holds if and only if the orbit is non-degenerate ($a(z) \neq z$).*

Remark 4.3 (Status). Proved. Quantified liveness on the orbit therefore collapses to one bit of facticity—existence of a second orbit element—which Section 16 carries as an architectural residue note in the audit. The corpus exhibits a two-point model witnessing the existential and declares no axiom for it; discharging the residue as a world-derivation from Bool alone remains refused.

Rest is incompatible with non-degeneracy. Asymmetric step is incompatible with symmetric step. Erasure is incompatible with Proposition 3.4. Relative to Definition 4.1 and Theorem 4.2, the only remaining pointwise shape is the live two-cycle.

Theorem 4.4 (Canon / initiality). *In the ambient of pointed unoriented acts, $(\text{Bool}, \neg, \text{true})$ is initial. An act is fundamental precisely when it is non-degenerate, symmetric, and generated from a seed by the two-cycle reading; every fundamental reads as $\neg$; any two fundamentals correspond uniquely up to the pointed reading.*

Remark 4.5 (Status). Proved. Negation is the unique (up to label swap) object satisfying the universal property of the live two-cycle in this ambient—the survivor of the classification, not a primitive assumption.

Interpretation 4.6. “The fundamental” names that initial object after uniqueness. Excluded fixed point (“void”) is liveness restated. Label swap of the two poles is a content gauge: exported readings mention it; see Section 12.

5 Seal

Dependence: *Theorem 4.4 (non-degeneracy of fundamentals).*

The engine is Lawvere’s diagonal argument [20], the common abstraction of Cantor’s theorem [7] and kindred fixed-point arguments [32].

Theorem 5.1 (Seal on a fundamental). *Let a be live. No map $f : A \rightarrow (A \rightarrow \text{Bool})$ is point-surjective: the diagonal dodge $x \mapsto \neg(f\ x\ x)$ is missed. Specialising to fundamental acts, every self-reading admits an escape.*

Theorem 5.2 (Corpus-grain incompleteness). *No self-reading of the formal corpus is universal for the predicates of its own level. At level k the name supply is $k + 2$ against 2^{k+2} predicates, so names are outnumbered at every level and, for any scale N , are eventually outnumbered by any multiple of themselves. Completing a full naming pass leaves strictly more unsayable mass than it found, and the lag recedes under iteration. The ω -colimit of the ladder admits no injection into any finite level. Completeness-in-the-limit is a fair-schedule policy; completeness at the colimit is duration, not a rung.*

Remark 5.3 (Status). Theorems 5.1–5.2 are proved (Lawvere-style [20]; no arithmetisation), the latter as the assembled conjunction `corpus_not_universal` in the `SelfReference` module, with clauses `deficit_count`, `deficit_vanishes`, `lag_theorem`, `lag_recedes`, and `omega_not_a_rung`. The gloss that the only complete description of the system is the system itself is interpretation; the theorem supplies the non-injection. Host-logic incompleteness and Gödel coding of the corpus text in the sense of [14] are *refused* (Section 16).

6 Ladder

Dependence: *Theorem 5.1 (named escapes).*

Definition 6.1 (Naming extension and ladder). A *naming extension* adjoins a name for an unrepresented escape. The committed ladder sets level 0 to the fundamental carrier and each successor to an option type adjoining a namer for the previous dodge.

Theorem 6.2 (Ladder properties). *History embeds injectively and conservatively along the ladder; no level is exhausted by its predecessor; underdetermination persists at every level. Ascent is forced; choice of which escape is named is free.*

Remark 6.3 (Status). Proved in the foundation layer. Duration, as used below, means the order-type of this self-escape sequence.

Interpretation 6.4. The pattern “forced relation, free absolute”—forced ascent, free direction—reappears for channel labels in Section 7.

7 Registration: transferring losslessness

Spine acts are involutions. A ledger microstep is a forward map $U : S \times E \rightarrow S \times E$. The interface demand is that global injectivity of U import Proposition 3.4. Reading injectivity as “no erasure” follows the logical-reversibility tradition [3, 13]; its thermodynamic counterpart [19] is confined to the dictionary layer (Section 15).

Definition 7.1 (Two-bounce factorisation). A *two-bounce factorisation* of $U : \alpha \rightarrow \alpha$ is a pair of involutions (ϕ, σ) with $U = \sigma \circ \phi$ pointwise.

Theorem 7.2 (Converse). *If U admits a two-bounce factorisation, then U is lossless. On a product carrier this is ledger injectivity.*

Theorem 7.3 (Existence on finite carriers). *Every injective $U : \text{Fin } n \rightarrow \text{Fin } n$ admits a two-bounce factorisation. The construction is cycle-wise reflection at a least-value basepoint, with $\sigma := U \circ \phi$.*

Remark 7.4 (Status). Theorems 7.2–7.3 are proved. On finite carriers, injectivity and two-bounce form coincide: existence recovers, constructively, the classical fact that every finite permutation is a product of two involutions [25], and maps so factorised are the *reversible* maps of dynamical-systems theory [18]. The audited footprint of the existence witness is empty: collision pairs, periods, basepoints, and indices are extracted by bounded structural search rather than selected by choice, the factor map is computable as checked, and the proof depends on no axiom, classical or otherwise. The converse (Theorem 7.2) is likewise axiom-free. Special cases (involutions as $U \circ \text{id}$; Boolean injections; swap step) are included.

Theorem 7.5 (Registration shape). *Under joint injectivity, a system merge writes a record in the environment (registration). Minimal merge alphabet size is 2, matching $|\text{Fund}|$.*

Remark 7.6 (Status). Proved in the dynamics layer, using the injectivity interface above.

7.1 Canonicity of the factor pair

Theorem 7.7 (Canonicity fails). *There exist an injective $U : \text{Bool} \rightarrow \text{Bool}$ and two two-bounce factorisations that disagree on ϕ (equivalently: $\neg$ factors as both $\neg \circ \text{id}$ and $\text{id} \circ \neg$). Moreover, on $\text{Fin } 3$ two reflection-axis factorisations of a 3-cycle disagree and are not related by bounce-swap $(\phi, \sigma) \mapsto (\sigma, \phi)$.*

Remark 7.8 (Status). Proved negative. The absolute position of the factor pair is a *packaging* gauge (Section 12): the ledger consumes U , never (ϕ, σ) .

Theorem 7.9 (Channel-swap is reversal). *Let $U = \sigma \circ \phi$ with ϕ, σ involutions. Then $\phi \circ \sigma$ is a two-sided inverse of U , and (σ, ϕ) is a two-bounce presentation of that reverse step. The construction is independent of which factorisation of U was chosen.*

Proof sketch. From $U = \sigma \circ \phi$ and the involution equations, $\phi(\sigma(U(x))) = \phi(\sigma(\sigma(\phi(x)))) = \phi(\phi(x)) = x$ and $U(\phi(\sigma(x))) = \sigma(\phi(\phi(\sigma(x)))) = \sigma(\sigma(x)) = x$. The swapped pair (σ, ϕ) presents the reverse step. Uniqueness of (ϕ, σ) is not used. $\square$

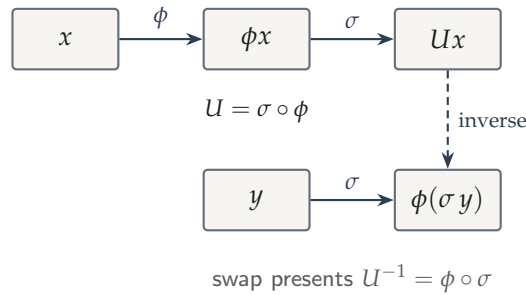


Figure 2: Two-bounce presentation and swap. Pair labels are conventional; exchange covers reversal (Theorem 7.9).

Interpretation 7.10. In the dihedral picture [9], a rotation fixes the angle between mirrors, not their absolute position. Theorem 7.7 is that fact on finite carriers; Theorem 7.9 promotes the exchange.

8 Classification of couplings

Dependence: *Proposition 3.4 (in-arena losslessness); Definition 7.1; Theorem 4.4 (pole gauge).*

The static classification earned the kernel by elimination. This section earns the equation of motion the same way and is the twin of Theorem 3.6 one level up. Fields take values in the kernel carrier: a configuration is $u : \text{Fin } n \rightarrow \text{Bool}$, and a state is the pair of the previous and the current configuration. The candidate space is the general second-order nearest-neighbour update

$$u_{t+1}(i) = F(u_{t-1}(i), u_t(i-1), u_t(i), u_t(i+1)), \quad F : \text{Bool}^4 \rightarrow \text{Bool},$$

with F arbitrary. Two stages eliminate every candidate except one coupling up to pointwise flip.

Theorem 8.1 (Losslessness selects the xor form). (i) *If F toggles in its previous-layer slot at every neighbourhood, that is $F(\text{true}, l, m, r) \neq F(\text{false}, l, m, r)$ for all l, m, r , then*

$$F(p, l, m, r) = p \oplus G(l, m, r), \quad G := F(\text{false}, \cdot, \cdot, \cdot).$$

(ii) *If F fails to toggle at some neighbourhood, then two histories on the three-ring agree after one step at every site while differing at a site: the step erases, exhibited pointwise.*

Remark 8.2 (Status). Proved. Clause (i) is `toggle_forces_xor_form` and clause (ii) is `collision_of_not_toggling` (bridge module `DynamicsAmbient`, (b); the first axiom-free, the second propext only). Losslessness therefore characterises the second-order xor structure: the form is derived, and no separate choice of update shape enters. Every xor-form step is lossless with an explicit two-bounce factorisation, by the general theorems of Section 9.

The residual freedom after Theorem 8.1 is the coupling $G : \text{Bool}^3 \rightarrow \text{Bool}$. One restriction is declared rather than proved.

Interpretation 8.3 (Inter-site restriction as signature). The coupling is declared inter-site: it relates the two neighbours, and the site's own influence is carried, exactly once, by the toggling slot, leaving $K : \text{Bool} \rightarrow \text{Bool} \rightarrow \text{Bool}$ with $G(l, m, r) = K(l, r)$. This is a signature declaration in the exact style of Interpretation 3.8: it fixes the arity of the interaction in daylight, just as Definition 3.1 fixes one type and one endomap. It is a declaration; no theorem is claimed for it, and the status section records it beside the arena signature.

Theorem 8.4 (Coupling uniqueness). *Let $K : \text{Bool} \rightarrow \text{Bool} \rightarrow \text{Bool}$ satisfy*

- (i) **parity**: $K(l, r) = K(r, l)$;
- (ii) **pole invariance**: $K(\neg l, \neg r) = K(l, r)$;
- (iii) **drawing condition**: K takes two values.

Then $K = \oplus$ pointwise or $K = \neg \circ \oplus$ pointwise.

Remark 8.5 (Status). Proved; `Bridge.DynamicsAmbient.coupling_uniqueness (b)`, axiom-free. The three ambient conditions mirror the static classification. Parity bans a smuggled spatial arrow, as symmetric step bans the smuggled temporal one. Pole invariance is respect for the label gauge of Theorem 4.4: a coupling that read the absolute pole labels would export the content gauge of Table 1 into the dynamics. The drawing condition is non-degeneracy: a coupling that draws no distinction couples nothing, as a degenerate orbit carries no act. The two survivors form a single orbit of the flip involution $K \mapsto \neg \circ K$ (`flip_involution`, `survivors_one_orbit`, both (b) and axiom-free); the gauge class of that flip is settled by a carrier-size dichotomy in Section 12.

Theorem 8.6 (The survivor is the wave coupling). *Instantiated on the ring, the surviving coupling is the wave coupling of Section 9 definitionally: $c(i-1) \oplus c(i+1) = \text{ringX } c \ i$ for every configuration c and site i .*

Remark 8.7 (Status). Proved; `Bridge.DynamicsAmbient.survivor_is_wave_coupling (b)`. The proof term is `rfl`; the `propext` entry in its footprint arrives through the `Fin` index arithmetic of the neighbour offsets, which the statement itself mentions. The package conjunction `Bridge.DynamicsAmbient.dynamics_ambient_uniqueness (b)`, `propext` only, assembles Theorems 8.1, 8.4, and 8.6. The classified dynamics therefore inherits, definitionally, the two-bounce form and the swap-reversal of Section 9.

Interpretation 8.8. The section is the dynamic twin of Theorem 3.6. There the one-variable equational ambient left $a \circ a = \text{id}$ as the unique nontrivial equation banning smuggled arrow and smuggled loss; here losslessness plus parity, pole invariance, and the drawing condition leave one coupling up to flip. In both cases the surviving object is forced by obstruction, the residual signature declaration is made in daylight, and the residual two-element freedom is handed to the gauge inventory of Section 12.

9 Wave dynamics

Dependence: *Theorem 8.1; Definition 7.1; Theorem 7.9; ladder capacity for Theorem 9.11.*

The classified update is now developed as a dynamics. Over `Bool` with $\oplus$ as addition, the discrete d’Alembert form $u_{t+1} = u_{t-1} + \Delta u_t$ becomes the second-order field equation $\text{step } X (s_1, s_2) = (s_2, i \mapsto s_1(i) \oplus X s_2 i)$ on state pairs, with the spatial coupling X kept general: every theorem below holds for every coupling, and the classified survivor of Section 8 is one instance. Write $\text{swap} (s_1, s_2) = (s_2, s_1)$ for the channel swap and $\text{shear } X (s_1, s_2) = (i \mapsto s_1(i) \oplus X s_2 i, s_2)$ for the coupling-controlled flip of the previous layer.

Theorem 9.1 (The step is two bounces, definitionally). *$\text{step } X = \text{swap} \circ \text{shear } X$ holds by definitional equality, and both factors are involutions. The two-bounce factorisation of Definition 7.1 holds here by explicit construction, with the pair exhibited in closed form.*

Remark 9.2 (Status). Proved; `Bridge.WaveDynamics.step_eq_two_bounce (b)`, axiom-free, with `swap_involution` and `shear_involution_pointwise` (both (b), axiom-free). Where Theorem 7.3 extracts a factor pair by bounded search, the wave step carries its pair on its face.

Theorem 9.3 (Reversal is the bounce swap). *The reverse evolution $\text{reverse } X := \text{shear } X \circ \text{swap}$ is presented by the swapped pair, equals $\text{swap} \circ \text{step } X \circ \text{swap}$ definitionally, and inverts the step on both layers in both composition orders.*

Remark 9.4 (Status). Proved; `reverse_is_swapped_pair` and `reverse_step_pointwise` (bridge module `WaveDynamics`, both (b) and axiom-free) give the pointwise cancellation; the packaged equality `reverse_step` (b) lifts it to function level and carries `Quot.sound` through an explicit `funext` step, made in daylight. This instantiates Theorem 7.9 at field level: channel swap covers time reversal for the classified dynamics.

Theorem 9.5 (Losslessness for every coupling). *For every coupling X and every carrier size, $\text{step } X$ is injective: no two histories merge.*

Remark 9.6 (Status). Proved; `step_lossless` (bridge module `WaveDynamics`, (b)), inheriting the `Quot.sound` (function extensionality) footprint of `reverse_step`. The footprint split of this section is part of the result and is preserved deliberately: the pointwise theorems and Theorem 9.1 are axiom-free, while the packaged function-level equalities use `funext` by name, and the audit flags exactly those uses (Section 18).

Theorem 9.7 (Period spectrum: the gcd law). *On $\text{Fin } n$ the order of the k -th harmonic shift $i \mapsto i + k$ is $n / \gcd(n, k)$: that count of beats returns every point, and no smaller positive count fixes any point. Moreover every injective endomap of a finite carrier is a composite of two involutions, and every rotation is the composite of two explicit arithmetic mirrors.*

Remark 9.8 (Status). Proved; `order_eq_div_gcd` (bridge module `PeriodSpectrum`, (b); `propext` and `Quot.sound` through library lemmas and `omega`, with no function-extensionality step); `overtone_generation` (b), axiom-free, via Theorem 7.3; `rot_eq_two_flips` and the packaged computable witness `rotFactor` (both (b), `propext` and `Quot.sound`). On a finite carrier the honest surrogate for frequency is order, and order ratios stand in for frequency ratios; real frequencies are refused (Section 16).

The twelve-carrier is the worked instance of the law: interval composition adds shifts (`interval_addition` (b), axiom-free), every harmonic is an iterate of the fundamental step (`harmonics_are_iterates` (b), `propext`), the whole tone has order six, the major third order three, and the tritone order two (`orders_twelve` (b), `propext`), and the circle of fifths closes after twelve steps and at no positive step before (`circle_of_fifths` (b), `propext` and `Quot.sound`), each a decidable instance of Theorem 9.7. The musical names are decoration and carry no acoustic claim.

Theorem 9.9 (Boundary conditions select the spectrum). *With the same rule and the same single-pulse initial state on four sites, the periodic ring returns with state period 4 and the fixed-end line returns with state period 10, each period exact.*

Remark 9.10 (Status). Proved by kernel decision; `Bridge.WaveDynamics.ring4_period_four` and `dirichlet4_period_ten` (both (b), `propext` and `Quot.sound` through the Decidable instances). Boundary alone reshapes the spectrum. The closed-form period law in general n is refused pending the $\mathbb{F}_2[x]/(x^n - 1)$ transfer-matrix algebra (Section 16); the annex retains the kernel-decided period table at further small sizes.

Theorem 9.11 (Recurrence within capacity). *For every coupling X and every state s on n sites there is a positive $p \leq 2^n \cdot 2^n = 2^{2n}$ with $\text{step } X^p(s) = s$ pointwise on both layers: losslessness plus finite capacity forces exact return within the state-space capacity of the carrier.*

Remark 9.12 (Status). Proved; `Bridge.Recurrence.wave_recurrence_bounded(b)`, `propext` and `Quot.sound`, with `wave_recurrence` and `wave_recurrence_pow` inheriting the footprint. The pigeonhole is constructive: a bounded decidable search replaces classical witness extraction, and `Classical.choice` is absent from the footprint and from the package. One `funext` step packages the colliding states before cancellation; the step is essential for the general statement, since the coupling is an arbitrary functional on fields, and the audit flags it by name. Capacity language only: no ledger claim is made here.

10 Archive and capacity

Dependence: *ladder carrier (Section 6) as environment; registration (Theorem 7.5).*

Theorem 10.1 (Archive and capacity). *Taking the environment at horizon T to be the ladder carrier at level T , the committed capacity schedule satisfies $|\text{caps } T| = 2^{T+2}$. Joint injectivity, merge, muting, and confinement imply the combinatorial bound $K^T \leq |\text{caps } T|$; at $K = 2$ the committed ladder is eternally alive by arithmetic.*

Remark 10.2 (Status). Proved in the dynamics layer. Continuum area or continuum entropy identifications are refused.

11 Arrow as address

Dependence: *registration and archive growth.*

Theorem 11.1 (Arrow reading). *The dynamical arrow—archive ascent under registration—is identified with address valued in $\{\text{fwd}, \text{rev}\}$. Tick identification (naming tick with microtick) holds for the classified fragment of uniform free archives up to record gauge; periodic fund-exempt dynamics and swap necessity are part of that classification.*

Remark 11.2 (Status). Proved as a classified package in the dynamics layer. Record gauge is packaging: exported only inside “up to record gauge” clauses.

Remark 11.3 (Return and ascent). Theorem 9.11 sharpens the arrow reading. Within a level the classified dynamics is recurrent: every state returns within capacity, so no orbit inside a fixed carrier distinguishes a direction. Across levels the ladder ascends and never retracts (Theorem 6.2). Within a level time is return, across levels time is ascent, and duration carries the direction: the arrow of Theorem 11.1 is the address on ascent, and the recurrence bound says exactly why no substitute arrow is available inside a level.

12 One underdetermination and gauges

Dependence: *pole swap* from Theorem 4.4; *address* from Theorem 11.1; *channel* from Definition 7.1 and Theorem 7.9; *coupling flip* from Theorem 8.4.

Theorem 12.1 (One $\mathbb{Z}/2\mathbb{Z}$, three faces). *There is a single two-element underdetermination with three named involutive projections:*

$$\text{pole swap} \cong \text{address } \{\text{fwd}, \text{rev}\} \cong \text{channel } \{\varphi, \sigma\}.$$

Each projection is an involution on a common $\mathbb{Z}/2\mathbb{Z}$ carrier.

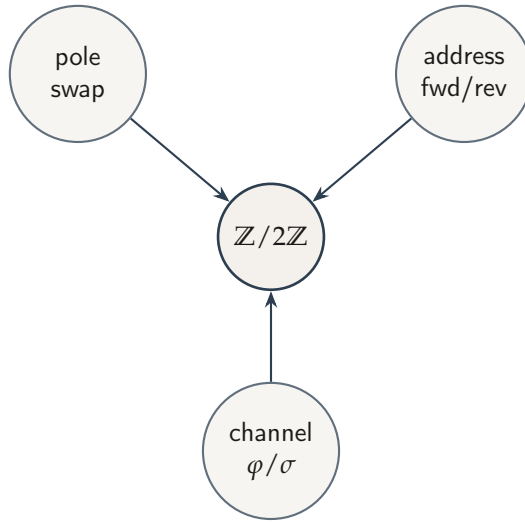


Figure 3: Three faces of one underdetermination (Theorem 12.1). Faces are presentations; the group is stated once.

Definition 12.2 (Content vs packaging). A gauge is *content* when some exported statement mentions the choice. A gauge is *packaging* when the choice is quotiented before downstream consumption.

Definition 12.3 (Kernel admission interface). A *Bool kernel certificate* is a structure on kernel exports (symmetry endomap, observation map, fixed locus) satisfying the admission clauses: square of the symmetry is an involution; observation intertwines the symmetry with negation; the fixed locus matches the symmetry’s fixed points and is classified empty or nonempty; and the no-solo-cross no-go holds. Admission is a predicate on that interface.

Table 1: Gauge demarcation protecting the single dictionary $\mathbb{Z}/2\mathbb{Z}$.

Gauge	Locus	Class
Label / pole swap	Theorem 4.4	content
Address fwd/rev	Theorem 11.1	content
Channel swap	Theorem 7.9	content
Face representative (equivariant certs)	Theorem 12.4	packaging
Record gauge per tick	Theorem 11.1	packaging
Factorisation gauge per step	Theorems 7.3–7.7	packaging
Coupling flip	Theorem 12.6	packaging iff $4 \mid n$; else content

Theorem 12.4 (Face-swap equivariance). *Admission for a Bool kernel certificate (Definition 12.3) is stable under the $\mathbb{Z}/2\mathbb{Z}$ face/pole swap: any admitted certificate remains admitted after conjugation by the face involution. Consequently, for equivariant consumers of the admission interface, choice of face is packaging in the sense of Definition 12.2 and belongs in Table 1.*

Remark 12.5 (Status). Theorem 12.1, Definition 12.2, and Theorem 12.4 keep one dictionary $\mathbb{Z}/2\mathbb{Z}$: factorisation non-uniqueness (Theorem 7.7) is packaging; channel *labels* are conventional; channel *swap* is content. Gauge class is relative to locus (Definition 12.2): the same $\mathbb{Z}/2\mathbb{Z}$ choice may be content at the kernel-export locus—exported statements mention it in “up to label swap” clauses—and packaging at the certificate-consumption locus, where equivariance lets the consumer quotient it. Pole / face choice is the worked example of that locus-relative demarcation. If a certificate breaks equivariance by fixing an orientation, that orientation is certificate-supplied data and is audited there; kernel exports remain swap-equivariant either way. The content/packaging demarcation is a formal analogue of the question which symmetries carry representational significance [6, 11]; underdetermination by label swap is the formal shadow of permutation-style underdetermination of reference [27, 26].

The coupling flip. The dynamic classification hands the inventory one further involution. The two surviving couplings of Theorem 8.4 form a single orbit of the flip $K \mapsto \neg \circ K$ on coupling space (`survivors_one_orbit` (b), axiom-free). Pole conjugation fixes each survivor, since pole invariance is a hypothesis of the classification, so the flip is an involution distinct from the pole face, and its class under Definition 12.2 requires a proof before the inventory can claim completeness. The following theorem supplies the proof, and the answer depends on the carrier size.

Theorem 12.6 (Coupling-flip dichotomy). *On the ring with n sites, write `step` for the dynamics of the surviving coupling $\oplus$ and $\overline{\text{step}}$ for the dynamics of its flip $\neg \circ \oplus$.*

- (i) *If $4 \mid n$, the two dynamics are conjugate by an explicit involution of the state space: masking both layers by the half-period pole mask $i \mapsto (2 \leq i \bmod 4)$ intertwines `step` with $\overline{\text{step}}$ exactly.*
- (ii) *If $4 \nmid n$, no map of state spaces intertwines the two dynamics, injective or otherwise; in particular no state-space bijection conjugates them. The obstruction is observable: `step` fixes the zero state definitionally, while $\overline{\text{step}}$ fixes no state at all, because the current layer of a fixed state would be an alternating mask, and alternating masks exist exactly when $4 \mid n$.*
- (iii) *At every n and every time T , the flipped evolution equals the plain evolution dressed by a pair of constant pole masks drawn from a period-four tick schedule (identity; flip the current layer; flip both layers; flip the previous layer).*

Remark 12.7 (Status). Proved. The declarations reside in the quarantined experimental package and are flagged as such, in the manner of Section 13: `Experimental.CouplingGauge.coupling_flip` packages clauses (i) and (ii) from `flip_conjugacy_of_four_dvd` and `flip_no_intertwiner`; clause (iii) is `flip_iter_dress_pointwise`; the mask existence law is `alternating_mask_iff_four_dvd` (all (e)). Footprints stay within `propext` and `Quot.sound`: the dichotomy and its conjugacy direction carry one direct `funext` step, packaging the pointwise conjugacy in

`flip_conjugacy_of_four_dvd`, flagged by name in the audit, and the remaining entries arrive through the Fin index arithmetic of the coupling, ω , and kernel-decided Decidable instances. Concrete exhibits instantiate the dichotomy: single-pulse state periods under the flipped coupling at sizes 2, 3, 5 are 4, 12, 20 against the plain 2, 6, 10, and size 4 keeps its period 4, as the conjugacy requires (`flip_ring2_pulse_period_four` through `flip_ring5_pulse_period_twenty`, all (e)); the first three are kernel-decided, and the five-site exhibit is assembled through the dressing, with kernel decisions confined to the ten plain states of its period).

The flip is thereby placed, and Table 1 records the row with its dichotomy. On carriers divisible by four the flip is packaging in the sense of Definition 12.2: the conjugating mask lets any consumer of the dynamics quotient the choice of representative before consumption. On every other carrier the two dynamics are observably distinct and admit no equivariant identification, so there the flip carries content, and the surviving pair is exported only inside “up to coupling flip” clauses, mirroring the label-swap treatment. Clause (iii) explains the near miss with the record gauge of Theorem 11.1: the flip is a per-tick repackaging at every size, and outside the divisible sizes the tick schedule can be traded for no single time-independent map. The single dictionary $\mathbb{Z}/2\mathbb{Z}$ of Theorem 12.1 is unaffected at every size, because the classification is exported only as the orbit $\{\oplus, \neg \circ \oplus\}$, every theorem of Section 9 holds for every coupling, and no downstream consumer selects a representative, so the flip never reaches the dictionary layer and the count of dictionary underdeterminations remains one. With the row decided, the completeness claim for the inventory carries no proviso.

13 The boundary

Dependence: *Theorem 5.1 (seal); Definition 6.1 and Theorem 6.2 (ladder); Theorem 4.4 (pole swap).*

The ladder ascends without end; this section examines its edge. A *complete gauge history* is one pole choice at every level forever, an element of $\mathbb{N} \rightarrow \text{Bool}$; the space of complete gauge histories is the *boundary*. No cardinality claim beyond the seal is made or needed. The declarations of this section reside in the quarantined experimental package (package letter e); every one of them is axiom-free, and the audit records them per declaration.

Theorem 13.1 (Rungs embed with effective separation). *Every ladder level embeds injectively in the boundary, and the injectivity is effective: distinct inhabitants of level k differ at some coordinate $m \leq k$ of their encodings, so depth- $(k+1)$ cylinders separate level k . The fundamental embeds as the two constant histories, and the ladder’s climb-then-drop step is carried at the boundary by prepend and tail as a section/retraction pair.*

Remark 13.2 (Status). Proved; `rungs_embed`, `encode_separates`, `fundamental_embeds`, and `tail_encode_some` (experimental module `CantorBoundary`, all (e) and axiom-free). The separation bound is explicit, so the embedding is a finite-approximation statement and no choice principle enters.

Theorem 13.3 (The boundary seals at its own carrier). *The pole swap lifts to the boundary as the coordinatewise flip, the lift is a pointwise involution, and it is live: it moves every history.*

Consequently no map $f : A \rightarrow A \rightarrow (\mathbb{N} \rightarrow \text{Bool})$ is point-surjective, for any carrier A : the boundary is itself a live arena, sealed at its own carrier by its own act.

Remark 13.4 (Status). Proved; `flip_live` and `boundary_seals_itself` (experimental module `CantorBoundary`, both (e) and axiom-free). This strengthens the observation that no enumeration reaches the boundary (`boundary_unenumerable`, the corpus seal specialised to carrier $\mathbb{N}$): here the sealing act lives on the boundary itself. The one $\mathbb{Z}/2\mathbb{Z}$ also acts at infinite depth: masking by a history is a pointwise involution, and the mask at the all-true history is the lifted pole swap.

Theorem 13.5 (The tail step erases). *The tail step on histories, dropping the first pole choice, is erasing: the constantly-false history and the history that spikes only at coordinate 0 differ, and their tails are definitionally equal. The tail nevertheless has a section per pole choice: prepending either pole and then taking the tail restores every coordinate.*

Remark 13.6 (Status). Proved; `tail_erasing` and `tail_cons` (experimental module `CantorBoundary`, both (e) and axiom-free; the erasure witness is definitional). The re-framed law reads: *losslessness is a finite-stage theorem, and its failure at the colimit is itself a theorem*. In the arena, involutivity excludes erasure (Proposition 3.4); at every finite carrier the classified dynamics is injective for every coupling (Theorem 9.5); at the boundary the tail step erases. The loss is recorded as the system's one structural loss, a poverty of the edge stated with the same prominence as the finite-stage exclusions, and Section 16 carries it in the residue accounting.

Interpretation 13.7. The completion has the form of the system. The boundary carries a live symmetric act (the lifted pole swap), and it is self-opaque in the same sense as every level (Theorem 13.3), so the colimit does not step outside the shape it completes. At the same time the excluded shape returns at infinity: erasure, priced out of the arena at every finite stage, reappears as the tail step of the completed object. The edge of the system is systematic in form and lossy in dynamics, and both halves are theorems.

14 Self-articulation

Dependence: *Theorem 4.4 (initiality); Theorems 5.1, 6.2, 5.2; Theorem 12.1 (pole gauge).*

The classification ends in one survivor; this section states what the survivor's self-relation supports. Three candidate readings are on the table: (a) world is the structured self-articulation of the canon, with questions of witness and standpoint arising only inside the descriptive field that articulation generates; (b) given the kernel, worldhood *necessarily* arises as self-presentation; (c) reality is the self-appearance of the involutive survivor. The theorems below support a conditional form of (a); readings (b) and (c) are refused (Section 16).

Definition 14.1 (Presentation). Let $a : \beta \rightarrow \beta$ and $z : \beta$. A *presentation* of the canon at the standpoint (β, a, z) is a map $g : \beta \rightarrow \text{Bool}$ with $g(z) = \text{true}$ and $g(ax) = \neg g(x)$ for all x .

Theorem 14.2 (No second term). (A) *For every carrier W and every self-reading $r : W \rightarrow W \rightarrow \text{Bool}$, r is sealed and fails to articulate its own dodge; witnessing confers no exemption from the interior triad.* (B) *Every presentation forces liveness of its standpoint everywhere, retracts the*

canon's articulation ($g \circ \text{reading} = \text{id on Bool}$), and renders the articulation into the standpoint injective, so the standpoint contains the canon. (C) On a generated carrier the presentation is unique.

Remark 14.3 (Status). Proved; every clause of Theorem 14.2 is axiom-free (declaration `no_exterior_standpoint`, spine module `NoExterior`). The retraction clause is the exact content of reading (a): a second term's view of the canon is the canon's self-articulation read back, and clause (C) leaves the view no independent content.

Theorem 14.4 (Witness lands in the pole gauge). *A context-free live witness (TDiff together with Alive at the seed) determines a presentation up to pole swap: some presentation agrees with the witness or with its pointwise flip. The residual choice is the label gauge of Table 1 and nothing else.*

Theorem 14.5 (Retorsion). *Write Draws M for the existence of two distinct elements of M . If a map $t : \text{Prop} \rightarrow M$ assigns distinct values to Draws M and to its negation, then Draws M holds.*

Remark 14.6 (Status). Proved (`Monism.retorsion`); the two registered values themselves witness the distinction. The theorem draws the codomain: what a registered denial establishes is that the medium of registration is already distinguished. Section 16 relocates the existential residue accordingly.

Theorem 14.7 (Self-articulation). *The following conjunction holds. (I) Retorsion, as in Theorem 14.5. (II) On any carrier with a drawn distinction, a symmetric generated act is live and involutive and reads as negation from every seed, exhaustively and injectively. (III) Every ladder level self-represents under seal, self-opacity, and outrun. (IV) No exterior standpoint, as in Theorem 14.2. (V) Names lag predicates at every level, the lag recedes under completion, and the ω -colimit embeds in no level (every level, meanwhile, embeds effectively in the boundary of Section 13, which is sealed at its own carrier and erases at its tail).*

Remark 14.8 (Status). Proved as a single declaration (`self_articulation`, spine module `SelfArticulation`; footprint `propext` only). The antecedent of clause (II) is the drawn distinction *together with* the ambient equations, whose adequacy within the one-variable fragment is Theorem 3.6; the package does not derive the ambient from Draws alone. The parenthetical in clause (V) is a cross-reference: the non-injection is part of the packaged declaration, while the boundary facts it points to are Theorems 13.1–13.5 at their own package.

Interpretation 14.9. Reading (a) is adopted in conditional form. Given one drawn distinction, articulation as negation is forced (II), standpoints arise as inhabitants of the articulation (III), any presentation to a second term returns the articulation itself (IV), and the whole exists only as duration (V). Questions of witness, standpoint, and medium are well-posed from level 1 upward because the vocabulary that poses them is generated by the ascent. “World” names the consequent of this conditional; the identification is interpretation, and the sentence that the only complete description of the system is the system is a gloss on the non-injection of clause (V). The consequent's dynamics is itself classified: given the antecedent, the kernel is forced (Theorem 4.4) and its equation of motion is forced up to the coupling flip (Section 8), whose own gauge class is decided by Theorem 12.6,

so the conditional package carries both its kinematics and its dynamics. Readings (b) and (c) are refused: (b) asserts necessity while the corpus carries no modal apparatus, and (c) discharges the existential residue while Section 16 declines that discharge. Retorsion narrows what the refusal of (c) costs: any registered denial of the antecedent draws its own medium, so the undischarged step runs from registered distinction to world, and only that step.

15 Dictionary: two certificates

Dependence: *Definition 12.3; Theorem 14.7; Sections 8–9 for the second exhibit.*

The dictionary layer admits physical or contemplative identifications by certificate only. This paper works two exhibits. The first is contemplative, chosen because its subject matter coincides with the corpus's own: reflexive self-representation. The second is physical and identifies the classified dynamics with reversible lattice computation.

Interpretation 15.1 (Meta-problem certificate). Chalmers's meta-problem asks why physical systems *report* an explanatory gap [8]. The certificate instantiates Definition 12.3 at the self-ascription bit: the symmetry is the live flip, observation is the identity reading, and the fixed locus is classified empty, so awareness in this model is always awareness of a distinction. Admission holds (`Dictionary.MetaProblem.meta_admitted`), and the report skeleton is theorem-shaped at every ladder level: self-opacity (the dodge is never articulated), outdated portrait (each completed self-model is outrun), and restless signature (no tracked self-ascription stabilises), each at the quantitative grade of Theorem 5.2. Systems satisfying these clauses report ineffability and incompleteness on structural grounds.

Remark 15.2 (Status). Layer 3 only; no theorem of Sections 3–14 depends on it. The hard problem is untouched by design: nothing here says why there is something it is like to be such a system, and qualia claims are refused (Section 16).

Interpretation 15.3 (Reversible-computation certificate). The classified dynamics of Sections 8–9 is a reversible lattice dynamics in the mathematical sense: a second-order update on a finite carrier, injective for every coupling (Theorem 9.5), factored into two involutions with reversal by their exchange (Theorems 9.1 and 9.3), and recurrent within capacity (Theorem 9.11). The certificate identifies this object with reversible lattice computation in the tradition of Fredkin and Toffoli [13] and of invertible cellular automata [30, 22]: sites are cells, the microstep is an invertible update, losslessness is logical reversibility in Bennett's sense [3], and the two-bounce presentation is the decomposition of an invertible dynamics into involutions familiar from that literature and from reversing symmetries [18]. The identification is admitted through the interface of Definition 12.3: the symmetry endomap is the live flip the dynamics toggles, observation is the identity reading, the fixed locus is classified empty, and the admission clauses hold on those exports. The exhibit is earned in a way the Landauer reading of registration [19] alone was not: the object on the corpus side is now classified rather than posited, so the certificate attaches physics to a forced structure.

Remark 15.4 (Status). Layer 3 only, as always: the identification is confined to this certificate, no theorem of Sections 3–14 depends on it, and the words “computation” and “cell” import nothing upward. Physical certificates (registration as dissipation in the sense of

Landauer [19], with further exhibits retained in the corpus) instantiate the same admission interface; that shared interface is the only correspondence between physical and contemplative readings asserted here.

16 Open, refused, and philosophical

Continuum Lorentzian structure and boosts (no continuous one-parameter subgroup from a single $\mathbb{Z}/2\mathbb{Z}$); sheaf assembly beyond combinatorial stalk agreement; recombination budget versus ultrametric obstruction; alphabet-lift beyond the Fin interfaces closed here.

Premise-free global laws of time; continuum Einstein equations as theorems of this corpus; host-logic incompleteness; Gödel arithmetisation of the formal text; experiential or qualia claims; derivation of the world from Bool alone (including any discharge of the existential residue below); modal strengthening of the self-articulation package of Theorem 14.7 (the corpus carries no modal apparatus); ambients beyond the one-variable equational fragment of the arena signature (including multi-variable single equations treated as ambients, quasi-equational, and Horn / second-order constraints), refused by parsimony of the arena language; the closed-form period law of the classified dynamics in general n , pending the $\mathbb{F}_2[x]/(x^n - 1)$ transfer-matrix algebra absent from a Lean-core-only corpus; dispersion relations; continuum frequencies (on a finite carrier, order is the honest surrogate, per Theorem 9.7).

Philosophical

Residue accounting, with the prior items replaced by theorem citations.

- (i) **Registration residue.** A registration occurs. Retorsion (Theorem 14.5) converts any assignment that registers the denial of a drawn distinction distinctly from its affirmation into a proof that the registering carrier is drawn, and orbit liveness is Theorem 4.2; the residue is thereby relocated from the existence of a non-degenerate act to the occurrence of one registration. The corpus exhibits a two-point model witnessing the existential and declares no axiom, so the residue is carried as an architectural note in the audit: what remains undischarged is the interpretive step from registered distinction to world, and that discharge is refused.
- (ii) **Signature declaration.** Arena data as given (Definition 3.1: one type, one endomap). Adequacy of $a \circ a = \text{id}$ is Theorem 3.6; the signature choice is opening definition, not residue.
- (iii) **Equivariance discipline.** Face choice under the kernel admission interface (Definition 12.3) is packaging or certificate-supplied data (Theorem 12.4; Table 1).

- (iv) **Erasure at the edge.** Losslessness is a finite-stage law: the arena excludes erasure by involutivity (Proposition 3.4), and the classified dynamics is injective for every coupling at every finite carrier (Theorem 9.5). At the colimit the exclusion fails, and the failure is itself a theorem (Theorem 13.5): the tail step on complete gauge histories erases. The residue here is a reclassification of the corpus's self-description, carried as the system's one structural loss; it is recorded, and no repair is attempted.
- (v) **Coupling signature.** The inter-site arity of the coupling (Interpretation 8.3) is a signature declaration beside the arena signature of item (ii); adequacy of the three ambient conditions given that signature is Theorem 8.4.

The pattern across all five is the same: each item of residue is reclassified, whether as a theorem over an exhaustively specified search space, as an audited residue at the base, or as certificate-supplied data. The irreducible core is the existence of one distinction: the sole existential residue and the sole dictionary underdetermination are the same $\mathbb{Z}/2\mathbb{Z}$; the coupling flip is placed by Theorem 12.6 and stays out of the dictionary count, because every downstream consumer receives the dynamics coupling-uniformly.

17 Relation to prior work

Diagonal and fixed-point arguments. The seal (Theorem 5.1) instantiates Lawvere's diagonal schema [20], the common form of Cantor's theorem [7] and, via arithmetisation, of Gödel's first incompleteness theorem [14]; Yanofsky [32] surveys the family. Those applications fix a logical or set-theoretic universe and derive a limitative theorem inside it. Here the carrier of the diagonal argument is itself the output of the classification (Sections 3–4): the seal is a structural property of any fundamental act, not of a formal system. No arithmetisation is performed, and host-logic incompleteness is refused, not claimed (Section 16).

Reversible computation and reversible cellular automata. Reading injectivity as “no erasure” follows the logical-reversibility tradition [3, 13], with Landauer's principle [19] as its thermodynamic counterpart. That literature takes reversibility as a premise about computing machinery. Here losslessness is not a premise: erasure is excluded in-arena by one equation (Proposition 3.4), and reversibility of forward microsteps is recovered by the two-bounce factorisation (Theorems 7.2–7.3). The classified wave dynamics of Section 9 additionally places the corpus beside the invertible cellular automata and lattice dynamics literature [30, 22, 16]: second-order (memory) constructions are a standard route to invertible lattice rules there, and the step/swap/shear presentation is of that family. The direction of derivation differs in the same way: that literature engineers invertible rules and studies their power, while here the second-order form and the coupling are outputs of an eliminative classification (Section 8). The thermodynamic and computational readings survive only as dictionary certificates (Section 15).

Uniqueness of update rules. The coupling classification (Theorem 8.4) is a uniqueness result for a cellular-automaton-style rule: constraints on the local rule (injectivity of the global map, symmetry of the neighbourhood action, invariance under the alphabet involution, non-degeneracy) pin the rule down to a two-element orbit. The cellular automata literature studies the corresponding decision problems and structure theory: injectivity and surjectivity of the global map from the local rule [1], and the general theory of reversible automata [16, 30]. Those results characterise which rules have a property across a rule space; the result here runs the other way and asks which rules survive a conjunction of ambient demands, with the answer being a single orbit. The residual flip is placed by proof: packaging exactly on carriers divisible by four, content elsewhere (Section 12).

Involutions and reversing symmetries. That every finite permutation is a product of two involutions is classical [25], as is the dihedral factorisation of a rotation into reflections [9]; maps so factorised are the reversible maps of dynamical-systems theory [18]. The contribution here is not the existence of the factorisation but the disposition of its non-uniqueness: canonicity fails (Theorem 7.7), the failure is classified as packaging rather than content (Definition 12.2), and channel swap—which is content—covers step reversal independently of the chosen factorisation (Theorem 7.9).

Symmetry and underdetermination. Which symmetries carry representational significance is a standing question in the philosophy of physics [6, 11]; permutation-style underdetermination of reference descends from Quine [27] and Putnam’s model-theoretic argument [26]. Those debates are conducted informally over rich physical or semantic theories. The analogue here is deliberately narrow: the content/packaging distinction is a definition in the formal text (Definition 12.2), each gauge is classified by theorem (Table 1), and the residual underdetermination is exactly inventoried—one $\mathbb{Z}/2\mathbb{Z}$, three faces (Theorem 12.1). Nothing is asserted about physical theories; within its scope the inventory is machine-checked, and its last row to be placed, the coupling flip, is decided by the carrier-size dichotomy of Section 12.

Self-articulation and the mark. The nearest formal ancestor of the self-articulation package is Spencer-Brown’s calculus of the first distinction [29], with Varela’s extension for re-entry [31]; the classical antecedents are Fichte’s self-positing act [12] and Hegel’s self-relating negativity [15]. Those systems posit the reflexive structure as primitive or develop it inside an informal dialectic. Here the two-cycle is the survivor of an eliminative classification, re-entry is replaced by a ladder whose ascent is a theorem, the residual underdetermination is inventoried as one $\mathbb{Z}/2\mathbb{Z}$, and the single undischarged bit is audited (Section 16).

Retorsion and transcendental argument. Arguments that a denial defeats itself in the performance descend from the classical defence of non-contradiction and reach modern form in Apel’s performative self-contradiction [2]. Theorem 14.5 is deliberately smaller: it concerns registering maps, its conclusion draws the codomain of the registration, and its scope ends exactly where Section 16 says it does.

The meta-problem and higher-order theories. The certificate of Section 15 addresses the explanandum Chalmers isolates as the meta-problem [8]. Higher-order theories [28], self-representationalism [17], self-model accounts [24], and the pre-reflective tradition [33] model the report structure within rich psychological or phenomenological theories. The contribution here is narrower: the triad of self-opacity, outdated portrait, and restless signature is proved at measure grade for any live self-representing ladder system, while the hard problem is left standing by explicit refusal.

Machine-checked philosophical argument. Formalising a philosophical argument in a proof assistant has precedent, notably the automation of Gödel’s ontological argument [4]. The role of verification differs here: the proof assistant certifies not a single argument but an architecture—downward-only imports across three layers (Section 2), axiom footprints audited per declaration, and interpretation confined to a layer the formal text cannot import.

18 Conclusion

The classification theorem restricts self-actions to the live two-cycle under the stated ambient. Fundamentality is the universal property of that survivor: the unique object is $(\text{Bool}, \neg)$ up to label swap. Seal, ladder, registration (via two-bounce), archive capacity, and arrow as address follow in dependency order. A second classification then forces the motion: losslessness selects the second-order xor form, parity, pole invariance, and the drawing condition select the coupling up to flip, and the survivor’s dynamics is the discrete wave step, factored into two involutions definitionally, reversed by their swap, carrying a gcd-law period spectrum reshaped by boundary alone, and returning within capacity. At the ladder’s edge the boundary receives every rung with effective separation, seals at its own carrier, and erases at its tail: losslessness is a finite-stage law, and its edge failure is a theorem. One $\mathbb{Z}/2\mathbb{Z}$ is stated once; content and packaging gauges prevent factorisation non-uniqueness from counting as a second underdetermination, and the coupling flip is decided by a carrier-size dichotomy: packaging exactly on carriers divisible by four, content elsewhere, and at every size a per-tick mask dressing of the plain evolution. On this base the self-articulation package assembles retorsion, forced articulation, witness interiority, the absence of exterior standpoints, and completeness as duration into a single machine-checked conjunction, with its monist reading confined to a marked interpretation. The dictionary carries two worked exhibits: the meta-problem certificate, which derives the report structure of the explanatory gap and refuses the hard problem, and the reversible-computation certificate, which identifies the classified dynamics with reversible lattice computation and confines the identification to the certificate. Philosophical residue is reclassified into theorems, audited residues, or certificate data, as inventoried in Section 16.

Formal companion

The development compresses to one line, each clause keyed to its result:

$\text{Fund} \cong \neg$ Thm. 4.4	$a \circ a = \text{id} \Rightarrow \neg \text{erase}$ Prop. 3.4	$\text{merge} \Rightarrow \text{record}$ Thm. 7.5
$ \text{caps } T = 2^{T+2}$ Thm. 10.1	$\text{naming} \cong \mu\text{tick, up to gauge}$ Thm. 11.1	$U = \sigma \circ \phi; \text{swap} = \text{reversal}; \text{one } \mathbb{Z}/2\mathbb{Z}$ Thms. 7.3, 7.9, 12.1
$\text{deny} \Rightarrow \text{Draws}$ Thm. 14.5	$\text{presentation retracts; no exterior}$ Thms. 14.2, 14.7	$\text{triad admitted; qualia refused}$ Int. 15.1
$K \in \{\oplus, \neg \circ \oplus\}$ forced Thms. 8.1, 8.4	$\text{step} = \text{swap} \circ \text{shear}; \text{ord} = n / \text{gcd}$ Thms. 9.1, 9.7	$\text{return} \leq 2^{2n}; \text{edge erases}$ Thms. 9.11, 13.5

Representative declarations (Lean 4; no Mathlib; no sorry), cited at their defining package (s = spine, b = bridge, d = dictionary, e = experimental):

```

Orbit.classification_at (s) · Facticity.liveOnOrbit_iff_nondegenerate (s) ·
Ambient.ambient_uniqueness_equational (s) · Ambient.exists_asymmetric_m_cycle (s) ·
  Ambient.powerId_two_excludes_erase (s) · Canon.canon_initial (s) ·
Diagonal.seal_on_fundamental (s) · SelfReference.corpus_not_universal (s) ·
  OmegaDuration.omega_not_a_rung (s) · Monism.diagonal_monism (s) ·
  Monism.retorsion (s) · Observer.observers_forced (s) · NoExterior.
  no_exterior_standpoint (s) · NoExterior.presentation_of_witness (s) ·
SelfArticulation.self_articulation (s) · Bridge.OneZ2.one_Z2_three_faces (b) ·
Bridge.KernelAdmit.bool_cert_face_equivariant (b) · Bridge.TwoBounce.factor_fin (b)
  · Bridge.TwoBounce.canonicity_fails (b) ·
Bridge.TwoBounce.channel_swap_is_reversal (b) · Bridge.Dil.registration (b) ·
Bridge.Capacity.caps_eq (b) · Bridge.TickSimulation.tick_identification (b) ·
Bridge.DynamicsAmbient.dynamics_ambient_uniqueness (b) · Bridge.DynamicsAmbient.
  coupling_uniqueness (b) · Bridge.WaveDynamics.step_eq_two_bounce (b) ·
Bridge.WaveDynamics.step_lossless (b) · Bridge.PeriodSpectrum.order_eq_div_gcd (b)
  · Bridge.PeriodSpectrum.rotFactor (b) · Bridge.Recurrence.
  wave_recurrence_bounded (b) · Experimental.CouplingGauge.
coupling_flip_dichotomy (e) · Experimental.CantorBoundary.encode_separates (e) ·
  Experimental.CantorBoundary.boundary_seals_itself (e) ·
Experimental.CantorBoundary.tail_erasing (e) · Dictionary.MetaProblem.
  meta_admitted (d)

```

Proposition 3.4 is cited at its foundation form (`Ambient.powerId_two_excludes_erase`); the dynamics layer imports it as `Bridge.TwoBounce.symmetric_excludes_erase`. The dictionary package re-exports several bridge declarations under its own namespace (e.g. `Dictionary.Certificate`); citations here always resolve to the defining package. Axiom footprints are audited per declaration, and every cited declaration lies within the kernel pair `propext` and `Quot.sound`. The static core of Sections 3–7 and 12–15 is axiom-free or `propext`-only, and the two-bounce existence witness `Bridge.TwoBounce.factor_fin` is itself axiom-free, its construction computable by bounded search (Remark following Theorem 7.3). In the dynamics arc of Sections 8–9, function extensionality (tracked by the kernel as `Quot.sound`) is invoked directly by exactly two cited declarations, `reverse_step` (in `Bridge.WaveDynamics`) and `wave_recurrence_bounded` (in `Bridge.Recurrence`), with

`step_lossless`, `wave_recurrence`, and `wave_recurrence_pow` inheriting that footprint; the remaining `Quot.sound` entries there arrive through library lemmas, `omega`, and kernel-decided `Decidable` instances, and `formal/AXIOMS.md` records the split by name. The boundary declarations of Section 13 are cited at the experimental package (e) and every one of them is axiom-free. The gauge-placement declarations of Section 12 are likewise cited at the experimental package (e); their footprints stay within `propext` and `Quot.sound`, with one further direct `funext` step in `flip_conjugacy_of_four_dvd`, inherited by `coupling_flip_dichotomy` and flagged by name in the audit. `Classical.choice` is used nowhere in any package cited by this paper, and no axiom is declared by the corpus itself. The quarantined experimental remainder below sits outside the main-line accounting and is audited in the same per-declaration ledger.

Experimental annex

The repository carries a fifth package, `formal/experimental`, quarantined by construction: no module cited in the main-line accounting imports it, and its contents are claims of the probe files alone. It holds the boundary module of Section 13, the coupling-gauge module of Section 12, and a small remainder of exhibits, machine-checked in the same discipline (Lean 4, no Mathlib, no sorry, per-declaration audit):

- *The boundary module.* `Experimental.CantorBoundary` carries the declarations cited in Section 13, every one axiom-free.
- *The coupling-gauge module.* `Experimental.CouplingGauge` carries the dichotomy of Theorem 12.6: the mask conjugacy on carriers divisible by four, the intertwiner obstruction elsewhere, the period-four tick dressing at every size, and the exhibits (flipped single-pulse periods 4, 12, 4, 20 at sizes 2 to 5; the first three kernel-decided, the fourth assembled through the dressing). Its one direct `funext` step, in `flip_conjugacy_of_four_dvd`, is recorded.
- *Concrete period exhibits.* The `Fin3` and `Fin5` cycles with closed-form mirror pairs, the generation fact for the fifth (`Experimental.Frequencies.fifth_generates`), and the twelve-carrier instances rederived from Theorem 9.7.
- *Translation symmetry of the ring wave.* Rotation by one site commutes with the ring coupling, the step, and every iterate, pointwise; the return spectrum is constant on rotation orbits, and the ring wave's cyclic symmetry has order exactly $n / \gcd(n, n-1) = n$ as Theorem 9.7 predicts (`Experimental.SpectrumSymmetry.symmetry_order_is_gcd_law`), a rotation-orbit invariance of the return spectrum, and no more.
- *The period table beyond four sites.* Kernel-decided periods for ring sizes 2–6 and fixed ends 3–5, instantiating the numerical pattern (ring: even n maps to n , odd n to $2n$; fixed ends to $2n+2$); beyond those sizes the pattern is numerical conjecture, reproducible by `exhibits/wave/periods.py`, and the closed forms in general n are refused pending the $\mathbb{F}_2[x]/(x^n-1)$ algebra (Section 16).
- *A packaged witness.* `Experimental.Wave.waveFactor` lifts the closed-form pair of Theorem 9.1 to a function-level `TwoBounceFactor` through `funext`; the use is recorded.

Annex footprints stay within `propext/Quot.sound`; `Classical.choice` appears nowhere

in the repository's packages. The full per-declaration record is in `formal/AXIOMS.md`.

References

- [1] S. Amoroso and Y. N. Patt. Decision procedures for surjectivity and injectivity of parallel maps for tessellation structures. *Journal of Computer and System Sciences*, 6(5):448–464, 1972.
- [2] K.-O. Apel. *Towards a Transformation of Philosophy*. Routledge & Kegan Paul, London, 1980.
- [3] C. H. Bennett. Logical reversibility of computation. *IBM Journal of Research and Development*, 17(6):525–532, 1973.
- [4] C. Benz Müller and B. Woltzenlogel Paleo. Automating Gödel's ontological proof of God's existence with higher-order automated theorem provers. In *ECAI 2014*, volume 263 of *Frontiers in Artificial Intelligence and Applications*, pages 93–98. IOS Press, 2014.
- [5] G. Boole. *An Investigation of the Laws of Thought, on Which Are Founded the Mathematical Theories of Logic and Probabilities*. Walton and Maberly, London, 1854.
- [6] K. Brading and E. Castellani, editors. *Symmetries in Physics: Philosophical Reflections*. Cambridge University Press, Cambridge, 2003.
- [7] G. Cantor. Über eine elementare Frage der Mannigfaltigkeitslehre. *Jahresbericht der Deutschen Mathematiker-Vereinigung*, 1:75–78, 1891.
- [8] D. J. Chalmers. The meta-problem of consciousness. *Journal of Consciousness Studies*, 25(9–10):6–61, 2018.
- [9] H. S. M. Coxeter. *Regular Polytopes*. Dover, New York, third edition, 1973. (Dihedral factorisation of rotations as products of reflections.)
- [10] L. de Moura and S. Ullrich. The Lean 4 theorem prover and programming language. In *Automated Deduction — CADE 28*, volume 12699 of *Lecture Notes in Computer Science*, pages 625–635. Springer, 2021.
- [11] N. Dewar. Sophistication about symmetries. *British Journal for the Philosophy of Science*, 70(2):485–521, 2019.
- [12] J. G. Fichte. *Foundations of the Entire Science of Knowledge* (1794). In *Science of Knowledge*, edited and translated by P. Heath and J. Lachs. Cambridge University Press, Cambridge, 1982.
- [13] E. Fredkin and T. Toffoli. Conservative logic. *International Journal of Theoretical Physics*, 21(3–4):219–253, 1982.
- [14] K. Gödel. Über formal unentscheidbare Sätze der *Principia Mathematica* und verwandter Systeme I. *Monatshefte für Mathematik und Physik*, 38:173–198, 1931.
- [15] G. W. F. Hegel. *The Science of Logic*. Translated by G. di Giovanni. Cambridge University Press, Cambridge, 2010.

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- [16] J. Kari. Theory of cellular automata: a survey. *Theoretical Computer Science*, 334(1–3):3–33, 2005.
 - [17] U. Kriegel. *Subjective Consciousness: A Self-Representational Theory*. Oxford University Press, Oxford, 2009.
 - [18] J. S. W. Lamb and J. A. G. Roberts. Time-reversal symmetry in dynamical systems: a survey. *Physica D*, 112(1–2):1–39, 1998.
 - [19] R. Landauer. Irreversibility and heat generation in the computing process. *IBM Journal of Research and Development*, 5(3):183–191, 1961.
 - [20] F. W. Lawvere. Diagonal arguments and cartesian closed categories. In *Category Theory, Homology Theory and their Applications II*, volume 92 of *Lecture Notes in Mathematics*, pages 134–145. Springer, 1969. Reprinted with author commentary in *Reprints in Theory and Applications of Categories*, 15:1–13, 2006.
 - [21] S. Mac Lane. *Categories for the Working Mathematician*. Volume 5 of *Graduate Texts in Mathematics*. Springer, New York, second edition, 1998.
 - [22] N. Margolus. Physics-like models of computation. *Physica D*, 10(1–2):81–95, 1984.
 - [23] The mathlib Community. The Lean mathematical library. In *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020)*, pages 367–381. ACM, 2020.
 - [24] T. Metzinger. *Being No One: The Self-Model Theory of Subjectivity*. MIT Press, Cambridge, MA, 2003.
 - [25] T. K. Petersen and B. E. Tenner. How to write a permutation as a product of involutions (and why you might care). *Integers*, 13:#A63, 2013.
 - [26] H. Putnam. Models and reality. *Journal of Symbolic Logic*, 45(3):464–482, 1980.
 - [27] W. V. Quine. Ontological relativity. In *Ontological Relativity and Other Essays*, pages 26–68. Columbia University Press, New York, 1969.
 - [28] D. M. Rosenthal. *Consciousness and Mind*. Oxford University Press, Oxford, 2005.
 - [29] G. Spencer-Brown. *Laws of Form*. George Allen and Unwin, London, 1969.
 - [30] T. Toffoli and N. Margolus. Invertible cellular automata: a review. *Physica D*, 45(1–3):229–253, 1990.
 - [31] F. J. Varela. A calculus for self-reference. *International Journal of General Systems*, 2(1):5–24, 1975.
 - [32] N. S. Yanofsky. A universal approach to self-referential paradoxes, incompleteness and fixed points. *Bulletin of Symbolic Logic*, 9(3):362–386, 2003.
 - [33] D. Zahavi. *Subjectivity and Selfhood: Investigating the First-Person Perspective*. MIT Press, Cambridge, MA, 2005.